Title: Elliptic-curve Signature Protocols: Toward Fault-injection Attack Resistance in Digital Identity Management

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Authors: L. Okafor, M. Zhang, L. Haddad (Institute for Advanced Computing Research)

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Abstract:

Fault-injection attack resistance remains a critical challenge for cryptographic hash functions, particularly in digital identity management. This paper proposes a framework built on elliptic-curve signature protocols that explicitly accounts for secure key exchange over lossy links. We evaluate the framework on power traces from an ARM Cortex-M4 platform and compare it against 3 baseline methods. Experimental results show that the proposed approach improves encryption throughput by 13.7% while keeping ciphertext overhead within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of cryptographic hash functions has changed how researchers and practitioners approach fault-injection attack resistance, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, elliptic-curve signature protocols have attracted sustained attention as a promising route toward solving fault-injection attack resistance in realistic settings such as digital identity management. Despite considerable progress, existing solutions for fault-injection attack resistance tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in security level in bits frequently come at the expense of encryption throughput, forcing designers to accept trade-offs that are poorly understood.

Our work is motivated by the growing demand for digital identity management, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of fault-injection attack resistance, a practical design built on elliptic-curve signature protocols, and an empirical evaluation on power traces from an ARM Cortex-M4 platform. Understanding secure key exchange over lossy links is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Hoffmann et al. (2017) introduced an early framework based on elliptic-curve signature protocols, reporting a 42.4% improvement in ciphertext overhead over previous baselines. Subsequent work by Fernandez et al. (2013) extended this line of research to digital identity management, although the evaluation was restricted to small-scale settings. A survey by Novak et al. (2011) categorised existing approaches to fault-injection attack resistance into three families and highlighted the lack of standard benchmarks. Researchers working on cryptographic hash functions have explored ML-DSA lattice signatures as an alternative, with Vasquez et al. (2024) noting that the approach scales poorly beyond 32 components. More recently, Novak et al. (2014) combined elliptic-curve signature protocols with ML-DSA lattice signatures, achieving a 27.7% reduction in overhead on power traces from an ARM Cortex-M4 platform.

Several studies have criticised the reliance on synthetic workloads, arguing that results on power traces from an ARM Cortex-M4 platform may not generalise to digital identity management. In contrast to these efforts, our work treats fault-injection attack resistance and secure key exchange over lossy links jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to fault-injection attack

resistance as a persistent obstacle for digital identity management. Despite this body of literature, no prior study has systematically compared elliptic-curve signature protocols and ML-DSA lattice signatures under identical experimental conditions.

3. Methodology

We formalise fault-injection attack resistance as a constrained optimisation task in which the objective is to maximise encryption throughput while keeping encryption throughput within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on elliptic-curve signature protocols, and a validation layer that incorporates ML-DSA lattice signatures. All experiments were conducted on power traces from an ARM Cortex-M4 platform, which provides 8 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies elliptic-curve signature protocols to reduce the effect of secure key exchange over lossy links, followed by a calibration step that stabilises key generation latency. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 3 baseline methods, including a conventional approach built on ML-DSA lattice signatures and two recent variants reported in the literature.

The proposed design improves encryption throughput by 13.7% relative to the strongest baseline and reduces key generation latency by 29.6%, a statistically significant difference ($p < 0.05$). A sensitivity analysis over 500 parameter settings shows that performance remains stable, with variation in key generation latency below 13.9%. Threats to validity include the limited diversity of power traces from an ARM Cortex-M4 platform and the possibility that deployment in digital identity management introduces conditions not captured in our setup.

4. Conclusion

This paper investigated fault-injection attack resistance through the lens of elliptic-curve signature protocols, and demonstrated that a carefully designed pipeline can improve encryption throughput by 13.7% on power traces from an ARM Cortex-M4 platform. The findings suggest that cryptographic hash functions will continue to benefit from closer integration of elliptic-curve signature protocols and ML-DSA lattice signatures, particularly in digital identity management. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of secure key exchange over lossy links. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for fault-injection attack resistance.